

DATA ANALYTICS PORTFOLIO PROJECT

Churn Analysis & Customer Intelligence

Identifying high-risk OTT subscribers from a multi-table relational dataset — and turning the findings into revenue-protecting action.



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Aspiring Data Analyst | SQL, Python & Power BI

1 What This Project Covers

1

The Business Challenge

In the hyper-competitive OTT space — Netflix, Hotstar, Prime Video — retention is the only path to survival. This project puts me in the seat of a Data Analyst tasked with spotting high-risk subscribers using demographics, subscription tiers, and support history.

2

Core Tech Stack

SQL & Python integration (numpy, pandas, sqlalchemy, matplotlib, seaborn), data cleaning, feature engineering, behavioral visualization, and writing insights an executive can act on.

3

Project Milestones

Relational data extraction across three linked tables · advanced feature engineering (tenure, churn rate, customer aging) · executive reporting that turns technical findings into business decisions.

Note: This isn't just a coding exercise — it's a business solution, portfolio-ready and applicable to E-commerce, SaaS, Fintech, and Adtech alike.

2 Churn Analysis — Roadmap



Each stage builds an analysis-ready view of the subscriber base — from raw relational tables to a report a growth team can act on.

3 Churn Definition by Business

Churn analysis is the process of figuring out why customers stop doing business with you. For this, we only need to look at three things:

Who
Identifying which customers have left, or are likely to leave

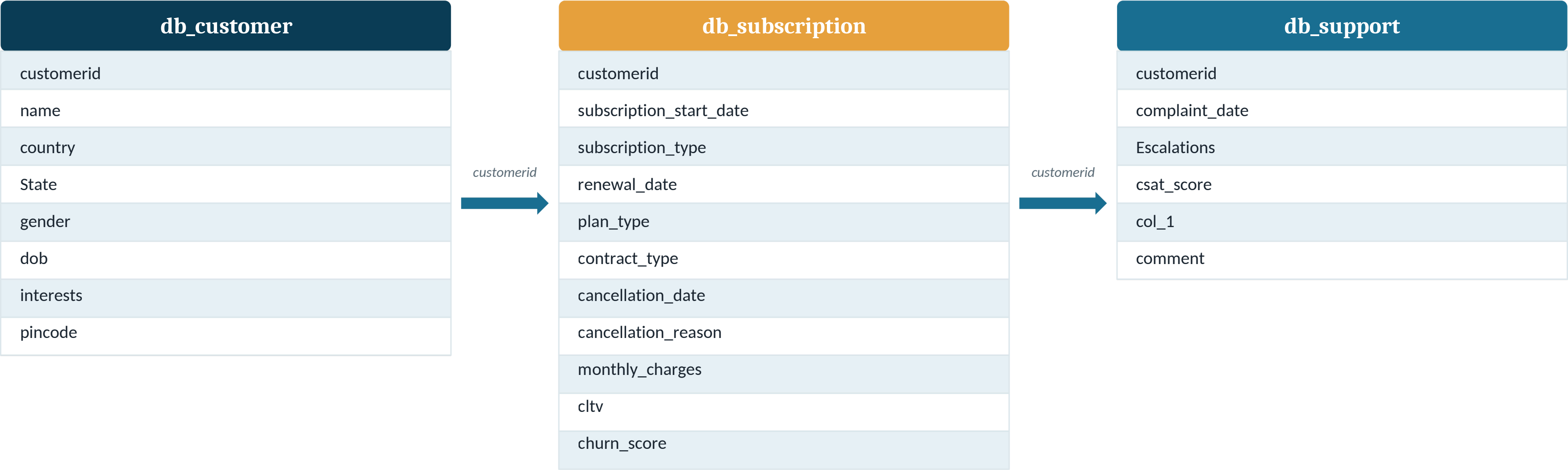
Why
Analyzing their behavior in the lead-up to cancellation

When
Finding the “danger zone” — the point where a user is most likely to leave

Business Type	Churn Definition
SaaS	Subscription cancelled
E-commerce	No purchase in 90 days
Adtech	No app/service use in 90–120 days
Streaming (this project)	Subscription cancelled
Telecom	Account terminated
Banking	No transactions for X months

4 Database Schema

Database : customer_churn | 1,300subscribers | 3 relational tables joined on customerid



5 Calculated Metrics

KPI	Formula
Churn Rate	Churned Customers / Total Customers
Retention Rate	1 – Churn Rate
Churn by Plan Type	Churn Rate GROUP BY plan_type (Basic / Standard / Premium / Enterprise)
Churn by State / Country	Churn Rate GROUP BY country, State
ARPU	SUM(monthly_charges) / COUNT(customerid)
Average Tenure	AVG(cancellation_date OR today – subscription_start_date)
Revenue at Risk	SUM(monthly_charges) WHERE churn_flag = 1
Escalation Rate	COUNT(Escalations = ‘Yes’) / COUNT(support tickets) × 100
Avg. Complaints / Customer	COUNT(complaints) / COUNT(DISTINCT customerid)
Escalation → Churn Correlation	CORR(escalations_flag, churn_flag)

6 Key Insights



Insights

- Basic-tier plans account for the least revenue impact from churn — most cancellations skew toward higher-value Premium and Business accounts (28.12% and 29.73% churn respectively), which is where the real revenue risk sits.
- Tamil Nadu carries both the largest customer base and the largest revenue exposure. Ontario shows the highest churn rate at 40%, on a small base worth monitoring rather than a confirmed trend.
- "Moved to a different location" is the single most common cancellation reason, followed closely by customers switching to a competitor.
- Escalations show almost no correlation with churn ($r = -0.0037$) — the churn-risk score (built from churn_score) is a far more reliable early-warning signal.

Action Items

- Investigate Ontario — pricing changes, complaint volume, or a local technical issue could explain the spike.
- Review whether Premium and Business-tier pricing or feature value recently shifted relative to competitors.
- Build a "found a better alternative" win-back flow — check what competitors are offering that we're not.
- Prioritize customers flagged High churn-risk: rank by CLTV, then reach out via email, SMS, and calls before they cancel.

7 Portfolio Project Summary

Resume-ready bullets, written from this project's actual results.

Churn & Revenue Impact

Engineered an end-to-end churn analytics pipeline for a 1,300-subscriber OTT platform by integrating three relational tables across demographics, subscription, and support data (12+ KPIs). Identified a 26.62% overall churn rate against 73.38% retention, quantified ₹338,693 in monthly recurring revenue already lost to churned accounts, and delivered a data-backed retention strategy targeting the highest-risk segments.

Risk Scoring & Segmentation

Built a churn-risk tiering model from churn_score, tenure, and plan data, segmenting the base into Low / Medium / High risk cohorts (High risk ≈ 23% of subscribers). Surfaced that Premium and Business-tier accounts churn at 28.12% and 29.73% respectively despite generating the most revenue per account, and recommended dedicated retention treatment for these cohorts over broad, undifferentiated offers.

Support Intelligence & Cancellation Analysis

Performed cross-functional support-churn correlation analysis by joining complaint, escalation, and CSAT data with subscription records. Found escalations carry almost no correlation with churn ($r = -0.0037$), and decomposed cancellation drivers into relocation, competitor switching, and pricing sensitivity — translating the findings into a prioritized, plain-language action list for a growth team.



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Thank you for reading — open to Data Analyst opportunities.